

Subject:
Behavioral Sciences

System:
Sensation, Perception, and Consciousness

Psychotic disorders are characterized by a loss of contact with reality. The most common psychotic disorder is schizophrenia, which has a typical onset of psychotic symptoms in early adulthood. Schizophrenia displays considerable inter-individual variation throughout the course of the disorder; many factors appear to influence its progression. Comorbid (co-occurring) conditions, common in schizophrenia, may lead to poorer prognoses. For example, research suggests that fewer depressive symptoms are moderately associated with higher rates of medication compliance, whereas more depressive symptoms are associated with increased future relapses. The early detection of psychosis also appears to have a significant impact on long-term outcomes in schizophrenia. A first psychotic episode that goes untreated for a longer period is associated with more future hospitalizations and a diminished response to future therapeutic interventions.

Researchers conducted a study to determine which was better at predicting the development of psychosis, a major component of schizophrenia, in high-risk young adults: the presence of certain speech pattern features or a standard clinical symptom inventory. The participant sample was restricted to 50 younger adults (age 18–30) who either met criteria for prodromal syndrome (preclinical signs of psychosis) or had a first-degree relative with schizophrenia.

Every three months for three years, participants were interviewed and given the Symptoms of Psychosis Inventory (SPI), a standardized clinical assessment designed to

measure psychosis and related symptoms. The SPI contains two subscales based on the major symptoms of schizophrenia: the Positive Symptom Scale (PSS) and the Negative Symptom Scale (NSS), which are summed to provide a total SPI score. Transcribed interview data were analyzed with software designed to assess the following speech pattern features: semantic coherence (the relatedness of phrases to one another), the number of determiners (eg, "that," "which," "what") used in a phrase (determiner density), and phrase length.

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Table 1 Comparison of Predictive Models

	Psychosis present (n = 11)	Psychosis absent (n = 39)
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Which of the following r values for the correlation between

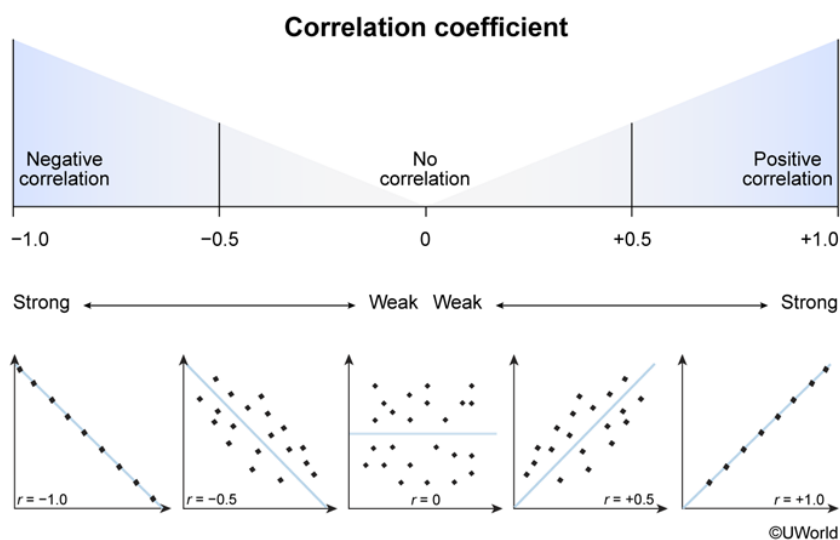
depressive symptoms and medication adherence in individuals with schizophrenia best supports the research presented in the first paragraph?

- ☐ A. +0.95 [14%]
- ☐ B. +0.30 [14%]
- ✓ ☒ C. -0.35 [44%]
- ☐ D. -0.90 [26%]

Correct

43% answered correctly

Explanation:



Comorbidity refers to the **simultaneous** presentation of **two or more psychological disorders**. If an individual with schizophrenia also has depression (a mood disorder), the two disorders are comorbid. Comorbid disorders often interact with each other, worsening the course of either or both. Disorders are thought to co-occur because they have common risk factors, such as genetic predisposition or environmental stressors.

A **correlation coefficient (r value)** describes the **linear** relationship between two variables; r values range from **-1 to 1** and describe the direction (sign) and **strength** of an association.

The **sign** of the r value indicates a positive or negative association, and the closer r is to -1 or 1, the stronger the relationship. Correlations describe **relationships** between variables, but it is not possible to determine if one variable *caused* the change in the other.

Fewer depressive symptoms were moderately associated with higher rates of medication compliance (ie, as depressive symptoms decrease, medication compliance increases); therefore, the research presented in the first paragraph is best supported by the *negative* and *moderate* r value of -0.35.

(Choices A and B) +0.95 and +0.30 reflect positive, not negative, correlations.

(Choices A and D) +0.95 and -0.90 reflect very strong, not moderate, correlations.

Educational objective:

The correlation coefficient (r value) describes a linear relationship between two variables. The r value ranges from -1 to 1 and reflects the strength and direction (positive or negative) of the linear association between these two variables. Comorbidity refers to the simultaneous presentation of two or more psychological disorders.

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Based on the symptoms of schizophrenia, which statement is accurate regarding the symptoms measured by the SPI?

- ✔ ☒ A. Incoherent speech would be assessed by the PSS. [47%]
- ☐ B. Delusions would be assessed by the NSS. [11%]
- ☐ C. Emotional flattening would be assessed by the PSS. [5%]

- ☐ D. Reduced need for sleep would be assessed by the NSS. [35%]

Correct

46% answered correctly

Explanation:

Schizophrenia symptoms	
Positive ("pathological excesses")	Negative ("pathological deficits")
• Hallucinations: False sensory perception (eg, hearing voices) in the absence of stimuli • Delusions: False belief (eg, "the neighbors are aliens") despite evidence to the contrary • Disorganized/incoherent speech • Disorganized/agitated behavior	• Apathy: A lack of motivation • Social withdrawal • Flat affect: Emotional flattening/monotone speech • Poverty (lack) of speech • Anhedonia: Inability to experience pleasure

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Psychotic disorders are severe psychological disorders characterized by a loss of contact with reality. Schizophrenia is the most common psychotic disorder, with a prevalence of about 1% in the United States.

Schizophrenia typically involves both **positive symptoms**, which are "pathological excesses" (eg, hallucinations, delusions, disorganized speech), and **negative symptoms**, which are "pathological deficits" (eg, apathy, inability to experience pleasure). Many people with schizophrenia also experience psychomotor symptoms (ie, changes in muscle tone or activity), which can occur either as a symptom of schizophrenia or as a side effect of medication.

Incoherent speech, a *positive symptom* of schizophrenia, would be assessed by the PSS (Positive Symptom Scale).

(Choice B) Delusions are a *positive* symptom of schizophrenia, so they would be assessed by the PSS (Positive Symptom Scale), not the NSS (Negative Symptom Scale).

(Choice C) Emotional flattening is a *negative* symptom of schizophrenia, so it would be assessed by the NSS (Negative Symptom Scale), not the PSS (Positive Symptom Scale).

(Choice D) Reduced need for sleep is a symptom of *mania*, which is also characterized by increased energy or agitation, flight of ideas (rapid thoughts), feelings of grandiosity, distractibility, and/or impulsive or reckless behaviors. Mania is indicative of bipolar disorder (not psychosis or schizophrenia), so it would not be assessed by the SPI.

Educational objective:

Schizophrenia is a psychotic disorder characterized by positive symptoms (eg, hallucinations, delusions, incoherent speech) and negative symptoms (eg, apathy, emotional flattening, social withdrawal).

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Which conclusion is best supported by the data in Table 1?

- ☐ A. The SPI Model resulted in fewer false positives than the Speech Model. [5%]
- ☐ B. The Speech model accurately predicted which 11 participants would develop psychosis. [5%]
- ☐ C. Clinical symptoms are less important than speech symptoms in diagnosing psychosis. [3%]
- ✔ ☒ D. The Speech Model correctly predicted more cases of psychosis than the SPI Model. [85%]

Correct

85% answered correctly

Explanation:

	Psychosis present (n = 11)	Psychosis absent (n = 39)	
Speech Model			Speech Model correctly predicted 10 cases
Psychosis predicted (n = 11)	10	1	
Psychosis not predicted (n = 39)	1	38	
SPI Model			SPI Model correctly predicted 7 cases (ie, 3 less than Speech Model)
Psychosis predicted (n = 16)	7	9	
Psychosis not predicted (n = 34)	4	30	

■ Correct predictions of psychosis

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When diagnosing, detecting, or predicting the development or presence of a medical condition, two types of errors are possible: A **false-positive** error occurs when the condition is predicted (or said) to be present but in reality is absent; a **false-negative** error occurs when the condition is predicted (or said) to be absent but in reality is present.

In the passage, the Speech Model correctly predicted 10 of the 11 cases of psychosis, whereas the SPI Model predicted only seven of the 11 cases. Therefore, the Speech Model

was a better predictor of psychosis than the SPI Model.

(Choice A) The **SPI Model** resulted in more (not fewer) false positives (nine) than the Speech Model (one).

(Choice B) Although the **Speech Model** predicted that 11 participants would develop psychosis, one of those did not do so (ie, there was one false positive).

(Choice C) The table reflects the ability of each model to predict the future occurrence or diagnosis of psychosis; the data therein do not reflect which symptoms are less (or more) critical to the diagnosis itself.

Educational objective:

The detection of a medical condition can result in two types of errors. A false positive occurs when the condition is said to be present but in reality is not. A false negative occurs when the condition is said to be absent, but in reality is present.

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Which of the following would support the claim that the SPI is a valid assessment?

- ☐ A. Scores are positively correlated from one administration to the next. [6%]
- ☒ B. Scores are positively correlated to an already established measure of psychotic symptoms. [63%]
- ☐ C. Scores are similar for the same patient, regardless of

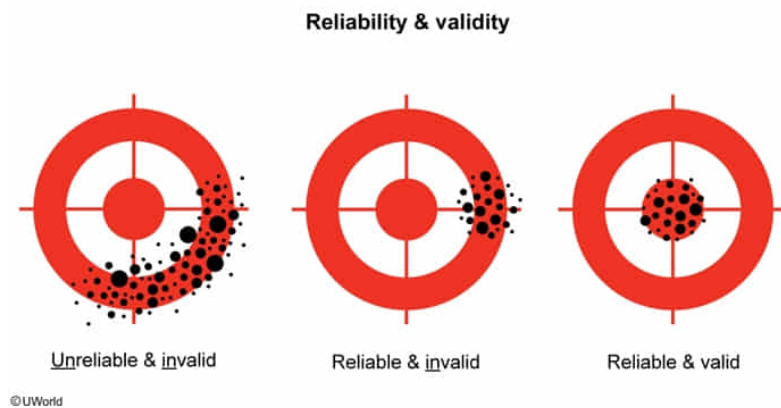
interviewer. [27%]

- ☐ D. Scores are similar for all participants at the end of the study. [2%]

Correct

63% answered correctly

Explanation:



A variety of criteria are used to judge the extent to which scientific findings are credible (ie, can be trusted or believed). Both the overall study design and particular measures used must be examined for reliability and validity.

Validity refers to the **accuracy** of a study or measure. **Internal validity** refers to the extent to which a measure or experiment produces a **true result**. **External validity**, also known as **generalizability**, is the extent to which study results can be **applied** outside the laboratory to real-life situations.

The validity of the SPI would be supported by scores that positively correlated to an already established measure of psychotic symptoms because when two different inventories that measure the same construct (eg, psychotic symptoms) are correlated, the validity (ie, accuracy) of both measures is supported.

(Choices A and C) Reliability refers to the extent to which an experiment or measure can consistently produce similar results every time. SPI scores that were positively correlated from one administration to the next and SPI scores that are similar regardless of interviewer both reflect two different types of *reliability* (not validity).

(Choice D) If SPI scores were similar for all participants, regardless of their diagnosis at the end of the study, this would *undermine* its validity; the SPI was designed to detect the presence of psychotic symptoms and should produce significantly different scores for people who later have a diagnosis of a psychotic disorder and those who do not.

Educational objective:

Validity refers to the accuracy of a measure or experiment (eg, a test accurately assesses what it was designed to measure).

Reliability describes the degree to which a measure or experiment is consistent (eg, a test produces a similar score for one person who takes it twice).

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Which of the following is accurate regarding the design of the study described in the passage?

- ☒

A. The null hypothesis is that speech pattern features and scores on the SPI will predict psychosis with similar accuracy. [41%]
- ☐

B. Speech pattern features are the dependent variables and the presence of psychosis is the independent variable. [14%]
- ☐

C. The results suggest that if a person exhibits prodromal symptoms, the person will eventually develop psychosis. [4%]
- ☐

D. The participants were assessed every three months for three years to monitor the progression and severity of psychotic symptoms. [39%]

Correct

42% answered correctly

Explanation:

		H ₀	
		True	False
Decision about H ₀	Reject	Type I error False positive: Falsely conclude there is a true difference	Correctly conclude there is a true difference
	Fail to reject	Correctly conclude there is no difference	Type II error False negative: Falsely conclude there is no difference

H₀ = null hypothesis (there is no true difference between two groups).
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A scientific hypothesis is a testable explanation for a phenomenon. The **alternative hypothesis** is based on prior evidence and assumes that a **significant relationship** or difference exists between variables (and often predicts the nature of that relationship or difference). In this study, the alternative hypothesis was that a significant difference *would* be found between statistical models of speech pattern features and clinical symptom assessments in the accuracy of psychosis prediction.

The inverse of the alternative hypothesis, the **null hypothesis**, states that there is **no significant difference** or relationship between the variables measured. Here, the null hypothesis was that no significant difference would be found between computer models of speech pattern features and scores on the SPI in the prediction of future psychosis.

Therefore, the null hypothesis is that speech pattern features and scores on the SPI will predict psychosis with similar accuracy.

(Choice B) This study was observational, so no true independent variable was manipulated by the researcher to establish a causal relationship. The speech pattern features and SPI measure are predictive variables assumed to indicate the risk of future psychosis.

(Choice C) All the study participants either met criteria for prodromal symptoms (preclinical signs of psychosis) or had a first-degree relative with schizophrenia. This study sought to determine whether speech pattern features or the SPI measure (not prodromal symptoms alone) could predict the development of psychosis.

(Choice D) Researchers were interested in determining whether certain speech patterns or a standard clinical inventory better predicted the development of psychosis, not the severity of disease.

Educational objective:

The null hypothesis states that no significant difference or relationship exists between variables, whereas the alternative hypothesis assumes that a significant relationship or difference does exist between variables.